Slomins VD05 Video Doorbell — Control4 Driver

Overview

Complete Control4 driver for Slomins VD05 Video Doorbell Camera with full CldBus API integration, MQTT event streaming, RTSP video streaming, doorbell functionality, and two-way audio support.

- **Model:** VD05
- **Version:** 6
- **Minimum Control4 OS:** 3.3.2+
- **Device Type:** Video Doorbell with Camera
- **Last Updated:** June 23, 2026

Features

Doorbell Functionality

- Doorbell ring detection and notifications
- Two-way audio communication (intercom)
- Visitor notification with snapshot attachment
- Integration with Control4 doorbell events
- Chime trigger support

Video Streaming

- RTSP streaming support (H.264/H.265)
- Main stream (high quality) and sub stream (low quality)

- Dynamic RTSP URL generation with token authentication
- Format: `rtsp://IP:8554/streamtype=0` (sub) / `streamtype=1` (main)
- Snapshot capture via HTTP API

Dynamic IP Auto-Update (Online Event Handling)

The driver includes intelligent handling of camera connectivity to ensure the correct IP address is always used.

Whenever the camera comes online, the driver automatically:

- Detects the **Camera Online** event
- Calls the **Get Devices API**
- Matches the device using **VID (Virtual ID)**
- Retrieves the latest **local IP address**
- Updates the **IP Address** property in the driver
- Pushes the updated address to the **Control4 Camera Proxy**

Real-Time Event Detection

The driver supports **MQTT-based real-time event streaming over SSL (port 8884)** and provides notifications for the following events:

- Doorbell ring notifications with snapshot
- Motion detection with snapshot attachment
- Human detection
- Stranger detection
- Camera online/offline status monitoring

Security & Authentication

- CldBus API integration with RSA-OAEP + SHA256 encryption
- HMAC-SHA256 signatures for API requests
- Secure token management (temp token, exchange token, auth token)
- MQTT over SSL/TLS (port 8884)
- OAuth-style authentication flow

- Pre-encrypted post data support for API calls

Network Discovery

- HTTP scan-based camera discovery (IP range scanning)
- SDDP (Simple Device Discovery Protocol) multicast discovery
- Automatic device binding to CldBus API
- Multi-device support via VID (Virtual ID)

Advanced Features

- Multiple connection types (Camera, Network/MQTT, Keep-alive TCP)
- Configurable event intervals (default: 5 seconds)
- Alert and info notifications (enable/disable)
- MQTT auto-connect and reconnection
- HTTP polling mode (alternative to MQTT)
- Custom authentication types: BASIC, DIGEST, NONE

Requirements

- **Control4 Composer Pro** (version compatible with OS 3.3.2+)
- **Control4 OS** 3.3.2 or newer (required for C4:Crypto() RSA encryption)
- **Network Access** to:
  - Doorbell on local LAN:
    - HTTP: 8080 (API, snapshots)
    - RTSP: 8554 (video streaming)
    - TCP: 3333 (device communication)
    - TCP: 8081 (keep-alive connection)
  - CldBus API: `https://api.arpha-tech.com`
  - MQTT Broker: Port 8884 (SSL/TLS)
- **Slomins VD05 Video Doorbell** with firmware supporting CldBus protocol

- **Account credentials** for CldBus API (default: pyabu@slomins.com)
- **Power:** Wired (doorbell transformer or AC adapter)

Device Specifications

VD05 Video Doorbell

- **Type:** Wired video doorbell camera
- **Power:** 16-24V AC (doorbell transformer) or POE
- **Resolution:** 1920x1080 (1080p), 1280x720 (720p), 640x480 (VGA)
- **Video Codec:** H.264, H.265
- **Field of View:** 180° wide angle
- **Night Vision:** IR LEDs
- **Audio:** Two-way audio (speaker + microphone)
- **Button:** Mechanical doorbell button with LED ring
- **Network:** Wi-Fi primary, Ethernet optional
- **Default IP:** DHCP assigned by router
- **VID:** Unique per device (obtained from CldBus API)
- **Ports:**
  - HTTP: 8080 (API, snapshots)
  - RTSP: 8554 (video streaming)
  - TCP: 3333 (device communication)
  - TCP: 8081 (keep-alive connection with auto-reconnect)
  - MQTT: 8884 (SSL/TLS events)

Installation

1. Install Driver Package

1. Download `Slomins-doorbell-VD05.c4z`

2. Open **Control4 Composer Pro**
3. Navigate to **System Design** > **Agents & Drivers**
4. Click **Add Driver** > **Browse** and select the `.c4z` file
5. Driver will appear as "Slomins VD05 Video Doorbell"

2. Add Doorbell to Project

1. In Composer Pro, drag "Slomins VD05 Video Doorbell" to your room (typically Porch/Entry)
2. The driver will auto-initialize and attempt connection
3. Check the driver properties for configuration

3. Configure Doorbell Settings

Navigate to driver **Properties** tab:

Required Settings:

- **IP Address:** Doorbell's local IP (e.g., assigned by your router via DHCP)
- **VID:** Doorbell's Virtual ID from CldBus API (unique per device)
- **Account:** CldBus account email (default: pyabu@slomins.com)

Optional Settings:

- **HTTP Port:** Default 8080
- **RTSP Port:** Default 8554
- **Authentication Type:** BASIC (default), DIGEST, or NONE
- **Username/Password:** If authentication required (default: SystemConnect/123456)
- **Enable MQTT:** True/False (default: False, enable for real-time doorbell events)
- **Enable Alert Notifications:** True/False (default: True)
- **Enable Info Notifications:** True/False (default: True)
- **Event Interval:** 5000ms (default, 0-30000ms range)

Advanced Settings:

- **Base API URL:** https://api.arpha-tech.com (default)
- **ClientID:** OAuth client ID

- **Public Key:** RSA public key (auto-populated)

4. Initialize Doorbell

Use the **Actions** tab to initialize:

1. **Login/Register:** Authenticate with CldBus API
2. **Get Temp Token:** Retrieve temporary token
3. **Get Exchange Token:** Exchange for persistent token
4. **Get Devices:** List all devices on account
5. **Bind Device:** Bind doorbell to Control4 driver
6. **Initialize:** Complete initialization sequence

Or use the single action: **Initialize** — Runs full initialization sequence automatically.

5. Test Streaming

Use **Actions** tab to test:

1. **Test Snapshot:** Verify snapshot URL generation
2. **Test Main Stream:** Test high-quality RTSP stream
3. **Test Sub Stream:** Test low-quality RTSP stream

Check the driver logs for RTSP URLs generated:

- Main: `rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=1`
- Sub: `rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=0`

6. Enable MQTT Events (Recommended for Doorbell)

For real-time doorbell ring notifications:

1. Set **Enable MQTT** property to **True**
2. Run action: **Get MQTT Info**
3. Run action: **Connect MQTT**
4. Verify connection in logs

Alternative: Use **HTTP Polling** for periodic status updates — Action: **Start HTTP Polling**

7. Configure Control4 Doorbell Integration

1. In Composer Pro, go to **Programming**
2. Create doorbell automation:
   - **When:** "Doorbell Ring" event
   - **Then:** Trigger chime, lights, notifications, etc.
3. Use **Doorbell Ring** event for Control4 announcements
4. Optionally forward video to TVs or mobile app

Usage

Doorbell Ring Event

When someone presses the doorbell:

1. VD05 detects button press
2. MQTT event sent to driver
3. Driver triggers Control4 **"Doorbell Ring"** event
4. Snapshot captured automatically
5. Notification sent with visitor snapshot
6. Control4 automation triggered (chime, lights, etc.)

Configure Doorbell Automation: Go to **Programming** > **When** > Select doorbell > **Doorbell Ring**, then add actions: Announce, send notification, turn on lights, etc.

Live Video Streaming

1. In Control4 app, navigate to doorbell camera
2. Tap camera to view live stream
3. Driver automatically generates RTSP URL with authentication token
4. View visitor at door in real-time

Manual RTSP Access:

- Main Stream: `rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=1`
- Sub Stream: `rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=0`

Two-Way Audio (Intercom)

Via Control4 App:

1. Open doorbell camera view
2. Tap microphone/speaker icon
3. Speak to visitor through doorbell speaker
4. Hear visitor through doorbell microphone

Note: Two-way audio requires Control4 app version that supports intercom on camera devices.

Event Notifications

When MQTT is enabled, driver receives real-time events:

- **Doorbell Ring** → Notification with snapshot attachment
- Motion detected → Alert
- Human/Stranger detected → Alerts
- Camera online/offline → Status updates

Configure in Properties:

- **Enable Alert Notifications:** Doorbell ring, motion, intruder alerts
- **Enable Info Notifications:** Online/offline, restart events

Camera Discovery

HTTP Scan Discovery:

1. Action: **Discover Cameras (HTTP Scan)**
2. Scans IP range (e.g., 192.168.1.1 to 192.168.1.50)
3. Set **IP Scan Range End** property (default: 50)
4. Finds doorbells on local network

SDDP Discovery:

1. Action: **Discover Cameras (SDDP)**
2. Sends multicast discovery packets
3. Doorbells respond with device info
4. Faster than HTTP scan

Snapshot Capture

Snapshot URL format:

```
http://[YOUR_DOORBELL_IP]:8080/tmp/snap.jpeg
```

Or use dynamic snapshot action: Action: **Test Snapshot** — Checks **Last Snapshot URL** property for result.

Properties Reference

Network Settings

Property	Type	Default	Description
IP Address	STRING	(DHCP assigned)	Doorbell's local IP address
HTTP Port	INTEGER	8080	Port for API and snapshots
RTSP Port	INTEGER	8554	Port for RTSP streaming
Authentication Type	LIST	BASIC	BASIC, DIGEST, or NONE
Username	STRING	SystemConnect	HTTP authentication username
Password	PASSWORD	123456	HTTP authentication password

Device Identification

Property	Type	Default	Description
VID	STRING	(unique per device)	Virtual device ID from CldBus
Product ID	STRING	VD05	Doorbell model identifier
Device Name	STRING	VD05	Friendly device name
Account	STRING	pyabu@slomins.com	CldBus account email

Streaming Settings

Property	Type	Default	Description
Stream Path	STRING	stream1	Legacy RTSP stream path
Snapshot URL Path	STRING	/GetSnapshot	Legacy snapshot path
Default Resolution	LIST	1280x720	640x480, 1280x720, 1920x1080

API & Authentication

Property	Type	Default	Description
Base API URL	STRING	https://api.arpha-tech.com	CldBus API endpoint
ClientID	STRING	(auto)	OAuth client ID
Public Key	STRING	(auto)	RSA public key from API
Temp Token	STRING	(auto)	Temporary authentication token
Exchange Token	STRING	(auto)	Exchange authentication token
Auth Token	STRING	(auto)	Persistent auth token

MQTT Settings

Property	Type	Default	Description
Enable MQTT	LIST	False	Enable real-time MQTT events

Property	Type	Default	Description
MQTT Host	STRING	(auto)	MQTT broker hostname
MQTT Port	STRING	(auto)	MQTT broker port (8884)
MQTT Client ID	STRING	(auto)	Unique MQTT client ID
MQTT Secret	PASSWORD	(auto)	MQTT authentication secret

Event Configuration

Property	Type	Default	Description
Event Interval (ms)	INTEGER	5000	Event polling interval (0-30000ms)
Enable Alert Notifications	LIST	True	Doorbell ring, motion, intruder alerts
Enable Info Notifications	LIST	True	Online/offline, restart events

Status Properties (Read-Only)

Property	Type	Description
Status	STRING	Driver initialization status
Online	STRING	Doorbell online status (true/false)
Main Stream URL	STRING	Generated main RTSP URL
Sub Stream URL	STRING	Generated sub RTSP URL
Last Motion	STRING	Last motion detection timestamp
Last Snapshot URL	STRING	Last captured snapshot URL
Last Clip URL	STRING	Last recorded video clip URL

Actions Reference

Initialization Actions

Action	Description
Initialize	Run full initialization sequence
Login/Register	Authenticate with CldBus API
Get Temp Token	Retrieve temporary token
Get Exchange Token	Exchange for persistent token
Get Devices	List all devices on account
Bind Device	Bind doorbell to driver

Streaming Actions

Action	Description
Test Snapshot	Test snapshot URL generation
Test Main Stream	Test high-quality RTSP stream
Test Sub Stream	Test low-quality RTSP stream

Discovery Actions

Action	Description
Discover Cameras (HTTP Scan)	Scan IP range for doorbells
Discover Cameras (SDDP)	Multicast SDDP discovery

MQTT Actions

Action	Description
Get MQTT Info	Retrieve MQTT broker info
Connect MQTT	Connect to MQTT broker

Action	Description
Disconnect MQTT	Disconnect from MQTT

Polling Actions

Action	Description
Start HTTP Polling	Start periodic status polling
Stop HTTP Polling	Stop HTTP polling

Utility Actions

Action	Description
Test Push Notification	Send test notification
Get Camera Properties	Retrieve doorbell properties
Update UI Properties	Update Control4 UI

Events

The driver triggers Control4 events for automation:

Event ID	Event Name	Description
1	Motion Detected	Doorbell detected motion (alert, with snapshot)
2	Doorbell Ring	Button pressed — primary doorbell event
3	Human Detected	Person identified in frame
4	Camera Online	Doorbell came online
5	Camera Offline	Doorbell went offline
6	Camera Restarted	Doorbell rebooted

Event Automation Examples:

Use Control4 programming to:

- Ring physical chime when doorbell pressed
- Announce "Someone is at the front door"
- Send push notification to mobile devices
- Turn on porch lights when doorbell rings
- Display video on TVs when motion detected
- Record video clip on doorbell ring

MQTT Configuration

Default Behavior

MQTT is **enabled automatically by the driver after authentication**. The driver will automatically:

1. Enable MQTT
2. Fetch MQTT credentials
3. Connect to the MQTT broker
4. Subscribe to camera event topics

You normally **do not need to enable MQTT manually**.

CldBus App Motion Detection Settings

To receive **Motion Detection** and **Human Detection** events, detection must be enabled in the **CldBus mobile app**.

Steps

1. Open the **CldBus App**

2. Select your **camera device**
3. Tap **Settings**
4. Navigate to `Settings → Motion Detection`
5. Under **Detection Type**, select one of the following:

Detection Type	Description
All Detections	All movement events will trigger notifications. This includes general motion detection.
Human Detection	Only human movement will trigger events. Non-human motion will be ignored.

Push Notification Configuration

1. Create Push Notification

1. Open **Composer Pro**
2. Navigate to `Agents → Push Notification`
3. Click **Add Notification** then enter a name for the notification.

Configure the following:

Setting	Value
Category	Cameras
Severity	Info / Critical
Subject	Camera Event

Click **Save**.

2. Enable Snapshot Attachment

Edit the created notification and set:

```
Attachment Type = Snapshot URL
```

This allows push notifications to include the **camera snapshot image**.

Mapping Push Notifications in Programming

1. In **Composer Pro**, open **Programming**.
2. Select the **Camera Driver** from the device list.
3. In the **Events** section, choose the event you want to trigger the notification (for example **Motion Detected**).
4. On the **right-side panel**, click **Push Notifications**.
5. From the dropdown list, select the **notification you created earlier**.
6. Drag and drop the notification into the programming area **or double-click the green arrow** to add it.

This maps the camera event to the push notification.

Notification Flow

```
Camera Event (MQTT)
        ↓
Driver Receives Event
        ↓
Driver Processes Event
        ↓
Control4 Event Triggered
        ↓
Push Notification Agent
        ↓
Mobile Notification Sent
        ↓
Snapshot Image Attached
```

Troubleshooting

Doorbell Not Ringing in Control4

Check:

1. MQTT is enabled and connected
2. **Enable Alert Notifications** is set to **True**
3. Doorbell button is functioning (test locally)
4. Event automation is configured in Control4 Programming

Action:

1. Run **Connect MQTT** to establish connection
2. Verify MQTT connection in logs
3. Press doorbell button and check driver logs for "Doorbell Ring" event
4. Configure Control4 automation for doorbell ring event

Doorbell Not Initializing

Check:

1. IP address is correct and doorbell is reachable
2. Account credentials are valid
3. VID matches the actual device
4. Doorbell is powered on and connected to network
5. Power supply is adequate (16-24V AC)

Action: Run **Initialize** action and check logs.

No Video Stream

Check:

1. RTSP port 8554 is open and accessible
2. Main/Sub Stream URL properties are populated
3. Network bandwidth is sufficient for streaming

Action:

1. Run **Test Main Stream** or **Test Sub Stream**

2. Check generated RTSP URL in logs
3. Verify network connectivity to doorbell

MQTT Events Not Working

Check:

1. **Enable MQTT** property is set to **True**
2. MQTT broker info is retrieved (Host, Port, Client ID, Secret)
3. Port 8884 is accessible
4. Network firewall allows MQTT over SSL

Action:

1. Run **Get MQTT Info** to populate MQTT settings
2. Run **Connect MQTT** to establish connection
3. Check **MQTT Host** and **MQTT Port** properties are filled
4. Verify in logs: `MQTT Connected`

Two-Way Audio Not Working

Check:

1. Control4 app version supports intercom on cameras
2. Audio permissions enabled on mobile device
3. Network has sufficient bandwidth for audio streaming
4. Doorbell microphone and speaker are functional

Action:

1. Test audio locally with Slomins app
2. Verify RTSP stream includes audio track
3. Check Control4 app settings for audio support

Snapshot Not Capturing

Check:

1. HTTP Port 8080 is accessible

2. Authentication credentials are correct
3. Snapshot path is valid

Action:

1. Run **Test Snapshot** action
2. Check **Last Snapshot URL** property
3. Test URL in browser: `http://[YOUR_DOORBELL_IP]:8080/tmp/snap.jpeg`

Discovery Not Finding Doorbell

HTTP Scan:

- Ensure doorbell IP is within scan range
- Adjust **IP Scan Range End** property
- Doorbell must respond to HTTP on port 8080

SDDP:

- Doorbell must support SDDP protocol
- Multicast must be enabled on network
- Check firewall allows multicast traffic

Authentication Failures

Check:

1. **Account** property has valid CldBus email
2. **Public Key** is retrieved from API
3. **Base API URL** is correct: `https://api.arpha-tech.com`
4. Network can reach API endpoint

Action:

1. Run **Login/Register** action
2. Check for "Authentication successful" in logs
3. Verify **Temp Token** property is populated

Doorbell Shows Offline

Check:

1. Power supply is connected and adequate (16-24V AC minimum)
2. Wi-Fi signal strength is sufficient
3. Network has not changed (SSID/password)
4. Doorbell hasn't been reset

Solution:

1. Check physical power connection
2. Verify Wi-Fi connection on doorbell
3. Run **Get Camera Properties** action
4. Check **Online** property in driver

Video Quality Issues

Check:

1. Network bandwidth is sufficient
2. Using appropriate stream (main vs sub)
3. Wi-Fi signal strength is good
4. Multiple devices not streaming simultaneously

Solution:

1. Use sub stream for lower bandwidth
2. Check network speed to doorbell
3. Move Wi-Fi access point closer if needed
4. Enable QoS for video traffic on router

Technical Notes

RTSP URL Format

The driver generates RTSP URLs in the format:

```
rtsp://[IP]:[PORT]/streamtype=[TYPE]
```

- **IP:** Doorbell IP address
- **PORT:** RTSP port (default 8554)
- **TYPE:**
  - `0` = Sub stream (low quality, low bandwidth)
  - `1` = Main stream (high quality, higher bandwidth)

Example URLs:

```
Main: rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=1
Sub:  rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=0
```

CldBus API Authentication Flow

1. **Initialize Camera:** GET `/v1/init` → Retrieve RSA public key
2. **Login/Register:** Encrypt account with RSA-OAEP-SHA256 → POST `/v1/LoginOrRegisterUser` → Get temporary token
3. **Get Temp Token:** Generate HMAC-SHA256 signature → GET `/v1/TempTokenGet` → Verify temp token
4. **Exchange Token:** POST `/v1/TempTokenExchange` → Get exchange token
5. **Bind Device:** POST `/v1/BindDeviceToUser` → Associate VID with account
6. **Ongoing API Calls:** Use `Authorization: Bearer [auth_token]` header with HMAC signature on each request

MQTT Event Format for Doorbell

Events received on MQTT topic: `{VID}/events`

Doorbell Ring Example:

```
{
  "event": "doorbell_ring",
  "timestamp": 1708300000,
  "snapshot_url": "http://[DOORBELL_IP]:8080/snapshot/12345.jpg",
  "device_id": "[YOUR_DEVICE_VID]",
```

```
  "button": "pressed"
}
```

Motion Detection Example:

```
{
  "event": "motion",
  "timestamp": 1708300000,
  "snapshot_url": "http://[DOORBELL_IP]:8080/snapshot/12346.jpg",
  "device_id": "[YOUR_DEVICE_VID]"
}
```

Driver parses events and triggers corresponding Control4 events.

Connection Types

The driver maintains multiple connections:

1. **Camera Connection (5001):** Type: CAMERA, Facing: 6 (bidirectional), Purpose: Control4 camera proxy interface, Hidden: True (doorbell button is primary interface)
2. **Network/MQTT Connection (6001):** Type: TCP, Ports: 3333 (API), 554 (RTSP), 8884 (MQTT SSL), Purpose: Network communication
3. **Keep-Alive Connection (7001):** Type: TCP, Port: 8081, Features: Auto-connect, keep-alive, connection monitoring, Purpose: Persistent connection to doorbell

Doorbell Button LED

VD05 has LED ring around button:

- **Solid White:** Normal operation, online
- **Pulsing White:** Processing (button pressed)
- **Red:** Offline or error
- **Blue:** Pairing/setup mode

Power Requirements

- **Voltage:** 16-24V AC
- **Current:** Minimum 1A recommended

- **Source:** Doorbell transformer or dedicated AC adapter
- **Note:** Insufficient power may cause random disconnections

Video Codec Support

- **H.264 (Recommended):** Widely compatible, lower bandwidth
- **H.265 (HEVC):** Better compression, higher quality, requires more CPU

Control4 prefers H.264 for maximum compatibility.

Integration Examples

Example 1: Doorbell Chime Automation

Scenario: Ring physical chime when doorbell pressed

```
When: Slomins VD05 → Doorbell Ring
Then:
  - Doorbell Chime → Activate
  - Wait 5 seconds
  - Doorbell Chime → Deactivate
```

Example 2: Video Display on TV

Scenario: Show doorbell video on living room TV when pressed

```
When: Slomins VD05 → Doorbell Ring
Then:
  - Living Room TV → Power On
  - Living Room TV → Select HDMI 1 (Control4 Matrix)
  - Control4 Matrix → Route Slomins VD05 to Living Room TV
  - Wait 30 seconds
  - Living Room TV → Return to Previous Input
```

Example 3: Security Mode Integration

Scenario: Enhanced alerts when away

```
When: Slomins VD05 → Motion Detected
If: Security System → State is "Away"
Then:
  - Send notification "Motion at front door while away"
  - Turn on: Porch Lights, Driveway Lights
  - Start recording: NVR Camera Group
  - Send snapshot to mobile app
```

Example 4: Package Delivery Detection

Scenario: Alert on human detection during day

```
When: Slomins VD05 → Human Detected
If: Time is between 9:00 AM and 6:00 PM
Then:
  - Announce "Delivery detected at front door"
  - Send notification with snapshot
  - Start recording for 2 minutes
```

Support

Slomins Support:

- Website: www.slomins.com
- Email: support@slomins.com
- Phone: Customer service hotline

CldBus API Issues:

- API Endpoint: https://api.arpha-tech.com
- Contact: API support via Slomins

Control4 Integration:

- Requires Control4 Composer Pro for installation

- Compatible with Control4 OS 3.3.2 and newer
- Doorbell automation programming available

Building the Driver

Files Structure

```
Slomins-doorbell-VD05/
├── driver.lua              # Main driver logic
├── driver.xml              # Configuration & metadata
├── mqtt_manager.lua        # MQTT event streaming & broker management
├── README.md               # This documentation
├── CldBusApi/              # API helper libraries
│   ├── auth.lua
│   ├── dkjson.lua
│   ├── http.lua
│   ├── sha256.lua
│   ├── transport_c4.lua
│   └── util.lua
└── build-c4z.ps1           # Build script
```

Build Package

Run in PowerShell:

```
.\build-c4z.ps1
```

This creates `Slomins-doorbell-VD05-v1.0.0.c4z` ready for installation.

Manual Build

```
$files = @(
    "driver.lua",
    "driver.xml",
    "mqtt_manager.lua",
    "README.md",
```

```
    "CldBusApi\*.lua"
)

Compress-Archive -Path $files -DestinationPath "temp.zip"
Rename-Item "temp.zip" "Slomins-doorbell-VD05.c4z"
```

Version History

v6.0 (Current)

- Fixed doorbell button response delay for programmed Control4 actions

v5.0

- Performance improvements and bug fixes

v4.0

- Removed PTZ capabilities

v3.0

- Implemented functionality to store all incoming events in the database.

v2.0

- Fixed version display showing 0 in Control4 Manage Drivers interface
- Fixed device specific conditionals labels not showing in Control4 Composer

v1.0.6

- Removed hardcoded AppName

v1.0.5

- Updated Control4 Camera Programming Options
- **Device Events**: Added "Anti Pry alarm is triggered", "Registered User Detected"

- **Commands Added**: Take Snapshot, Initialize Camera *, Unmute Mic, Mute Mic, Turn up speaker volume, Turn down speaker volume, Sensitivity =
- **Commands Removed**: TEST_SNAPSHOT, TEST_MAIN_STREAM, TEST_SUB_STREAM, BIND_DEVICE, INITIALIZE_CAMERA, LOGIN_OR_REGISTER, GET_TEMP_TOKEN, GET_EXCHANGE_TOKEN, GET_DEVICES, AWAKE_CAMERA, GET_MQTT_INFO, CONNECT_MQTT, DISCONNECT_MQTT, GET_CAMERA_PROPERTIES, UPDATE_UI_PROPERTIES, START_HTTP_POLLING, STOP_HTTP_POLLING, PRESET_SAVE, PRESET_RECALL
- **Conditions Added**: If sensing motion, If not sensing motion, If Mic is muted, If Mic is unmuted, If Speaker volume is 1-10, If battery is <,> or =, Sensitivity =

v1.0.4

- Bug fixes for low battery alert implementation, including timer and recovery logic

v1.0.3

- Send Control4 notification marked as critical if battery percentage is less than 15%
- Send notification every 4–6 hours (interval should be configurable in properties; default is 6 hours)
- Disable the timer if battery level is greater than 80%
- Enable the timer only when the first notification below 15% is received

v1.0.2

- convert the readme.md file to readme.html file

v1.0.1

-

v1.0.0

- Initial release
- CldBus API integration
- RTSP streaming support
- Doorbell ring detection

- Two-way audio support
- Motion and human detection
- MQTT event streaming
- HTTP polling mode
- Snapshot capture
- PTZ controls (digital zoom)
- Control4 event integration

License

Appendix: Quick Reference

Default Ports

- **HTTP:** 8080
- **RTSP:** 8554
- **TCP:** 3333, 8081
- **MQTT:** 8884 (SSL)

Example Device

- **Model:** VD05
- **IP:** Assigned by your router (DHCP)
- **VID:** Unique per device

RTSP URLs

```
Main: rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=1
Sub:  rtsp://[YOUR_DOORBELL_IP]:8554/streamtype=0
```

Snapshot URL

```
http://[YOUR_DOORBELL_IP]:8080/tmp/snap.jpeg
```

Primary Events

- **Doorbell Ring** — Button pressed
- **Motion Detected** — Movement at door
- **Human Detected** — Person identified
- **Stranger Detected** — Unrecognized person identified
- **Camera Offline** — Connectivity lost

End of Documentation